Wave Basics: Conceptual Questions

SYNOPSIS

Seventeen questions on what Notes 1 and 1a have just defined — what needs a medium and what does not, which way the medium moves, what a wave carries and what it leaves behind, and the difference between a pulse and a periodic wave. Nothing here is new material. If a question looks like one you have already answered, it probably is: several of them ask the same idea from the other side, which is the point.

Underline your answer for each situation.

PERFORMANCE EXPECTATIONS

Describe mechanical waves and how mechanical waves carry energy (but not materials). Describe, compare, and contrast transverse and longitudinal waves and give examples of each.

Conceptual questions

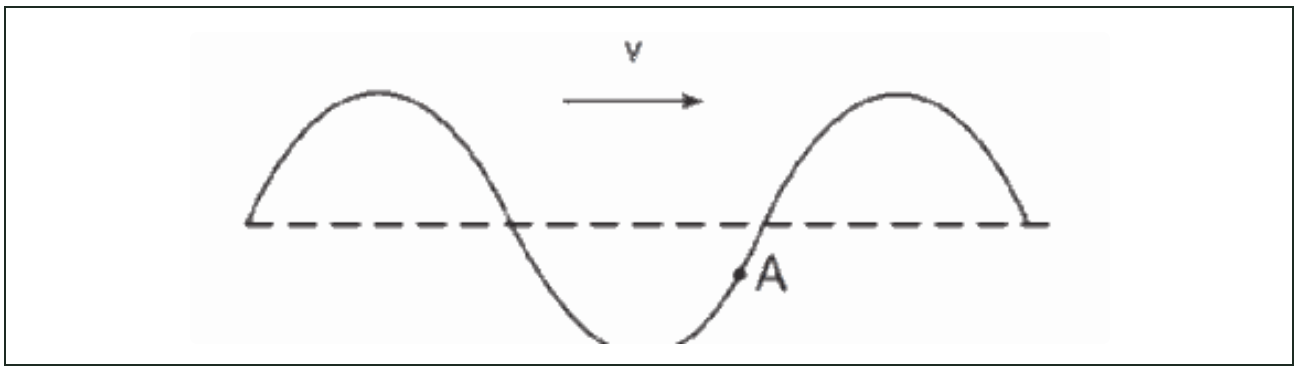
Underline your answer for each situation.

1. Which type of wave requires a material medium through which to travel?
    1. sound
    2. television
    3. radio
    4. x ray

2. A single vibratory disturbance moving through a medium is called
    1. a node
    2. an antinode
    3. a standing wave
    4. a pulse

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3. The diagram below represents a transverse wave traveling to the right through a medium. Point A represents a particle of the medium.



In which direction will particle A move in the next instant of time?

1. up
2. down
3. left
4. right

4. As a transverse wave travels through a medium, the individual particles of the medium move
   1. perpendicular to the direction of wave travel
   2. parallel to the direction of wave travel
   3. in circles
   4. in ellipses

5. A periodic wave transfers
   1. energy, only
   2. mass, only
   3. both energy and mass
   4. neither energy nor mass

6. Which type of wave requires a material medium through which to travel?
   1. radio wave
   2. microwave
   3. light wave
   4. mechanical wave

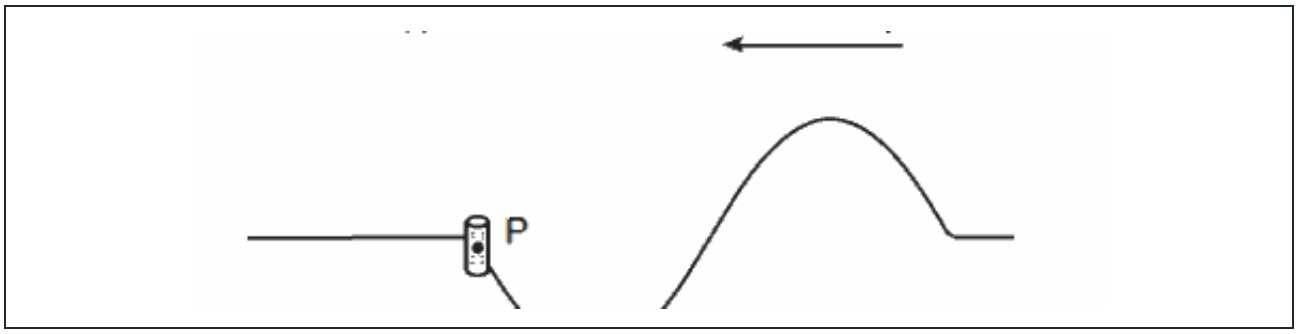
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7. A ringing bell is located in a chamber. When the air is removed from the chamber, why can the bell be seen vibrating but not be heard?
   1. Light waves can travel through a vacuum, but sound waves cannot.
   2. Sound waves have greater amplitude than light waves.
   3. Light waves travel slower than sound waves.
   4. Sound waves have higher frequencies than light waves.

8. Which statement correctly describes one characteristic of a sound wave?
   1. A sound wave can travel through a vacuum
   2. A sound wave is a transverse wave
   3. The amount of energy a sound wave transmits is directly related to the wave's amplitude
   4. The amount of energy a sound wave transmits is inversely related to the wave's frequency

9. A television remote control is used to direct pulses of electromagnetic radiation to a receiver on a television. This communication from the remote control to the television illustrates that electromagnetic radiation
   1. is a longitudinal wave
   2. possesses energy inversely proportional to its frequency
   3. diffracts and accelerates in air
   4. transfers energy without transferring mass

10. The diagram below represents a transverse water wave propagating toward the left. A cork is floating on the water's surface at point P.

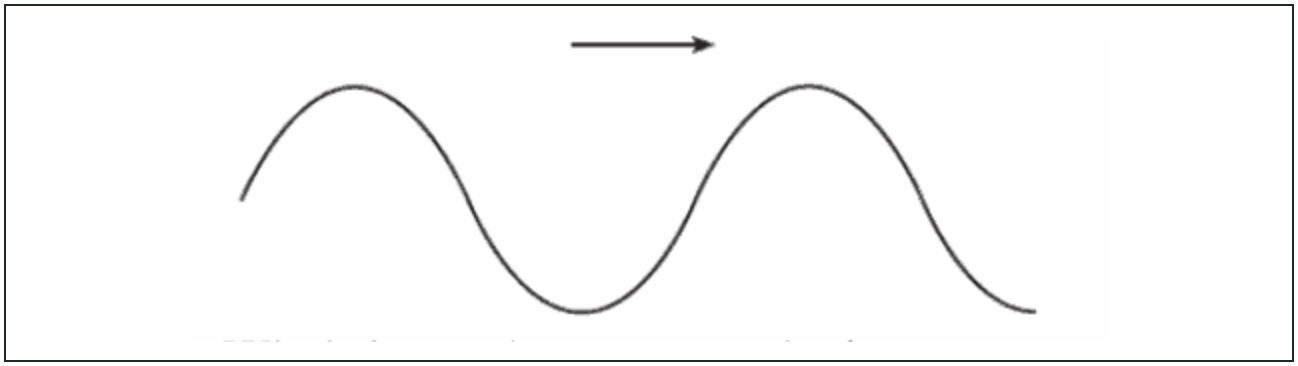


In which direction will the cork move as the wave passes point P?
   1. up, then down, then up
   2. down, then up, then down
   3. left, then right, then left
   4. right, then left, then right

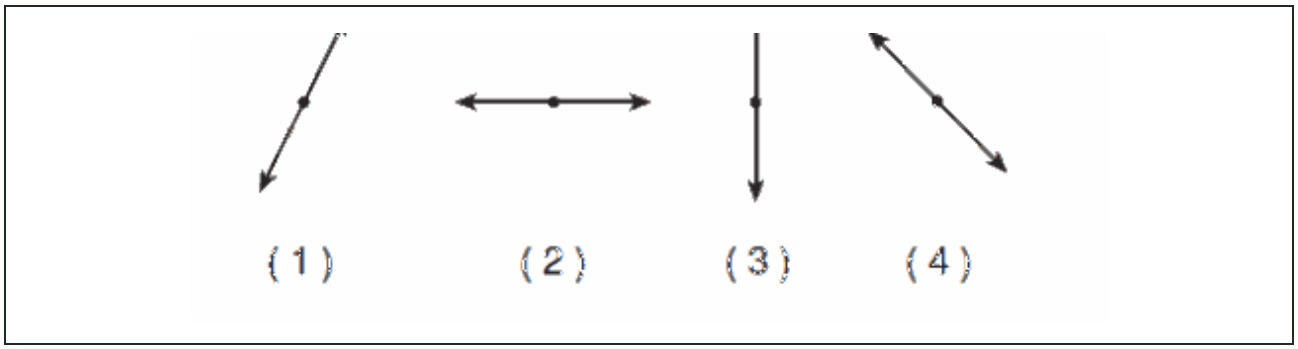
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11. A pulse traveled the length of a stretched spring. The pulse transferred
    1. energy, only
    2. mass, only
    3. both energy and mass
    4. neither energy nor mass

12. A transverse wave passes through a uniform material medium from left to right, as shown in the diagram below.



Which diagram best represents the direction of vibration of the particles of the medium?



13. A tuning fork vibrating in air produces sound waves. These waves are best classified as
    1. transverse, because the air molecules are vibrating parallel to the direction of wave motion
    2. transverse, because the air molecules are vibrating perpendicular to the direction of wave motion
    3. longitudinal, because the air molecules are vibrating parallel to the direction of wave motion
    4. longitudinal, because the air molecules are vibrating perpendicular to the direction of wave motion

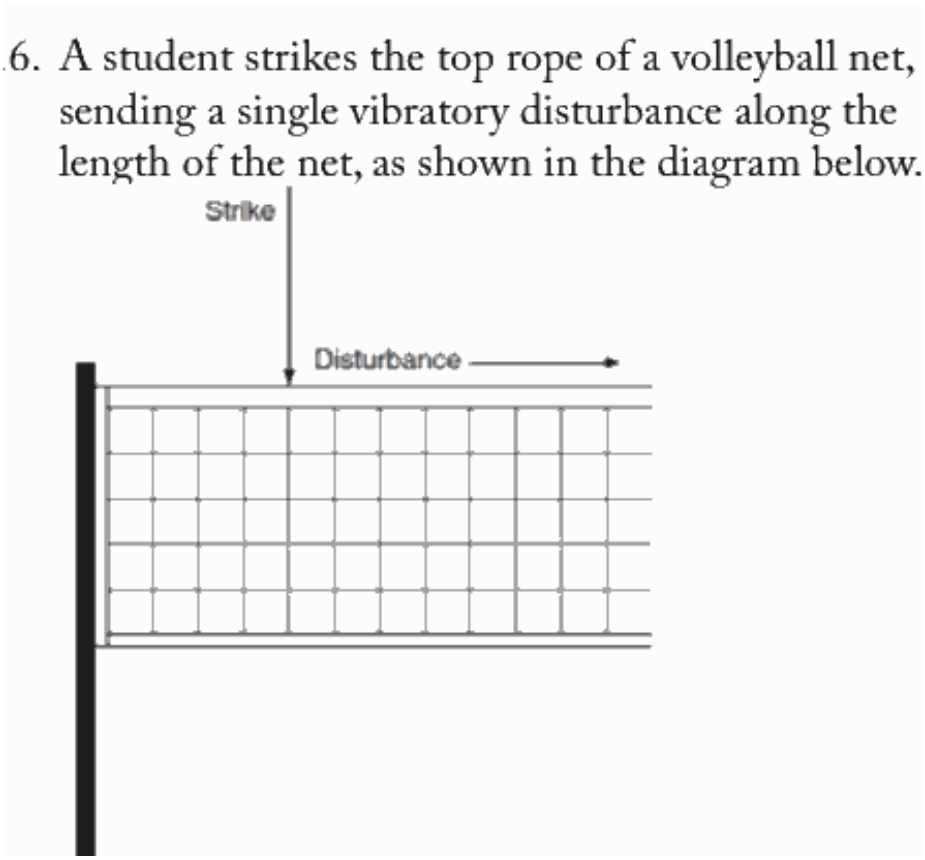
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14. Which form(s) of energy can be transmitted through a vacuum?
    1. light, only
    2. sound, only
    3. both light and sound
    4. neither light nor sound

15. How are electromagnetic waves that are produced by oscillating charges and sound waves that are produced by oscillating tuning forks similar?
    1. Both have the same frequency as their respective sources.
    2. Both require a matter medium for propagation.
    3. Both are longitudinal waves.
    4. Both are transverse waves.

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16. A student strikes the top rope of a volleyball net, sending a single vibratory disturbance along the length of the net, as shown in the diagram below.



This disturbance is best described as

1. a pulse
2. a periodic wave
3. a longitudinal wave
4. an electromagnetic wave

17. Which diagram below does not represent a periodic wave?



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Looking ahead

That is the end of the measuring-and-naming half of Module 1. **Note 2** stops reading things off rulers and asks what force makes a single bead oscillate at all — and the sines and cosines that Note 1a's synopsis promised come out of the answer rather than being asserted.

Where this sits. Three ideas from these seventeen do most of the work later in the module: a wave carries energy and not material, the medium's motion is perpendicular to travel for a transverse wave and parallel for a longitudinal one, and a pulse is not periodic so it has no frequency and no wavelength.

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